

Strategic Under-Specification: A Stackelberg Game Between User and Assistant

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Abstract

Users of coding assistants routinely under-specify their intent, and the assistant’s policy for handling missing information—guess silently, or ask a clarifying question—is itself an endogenous strategic choice that users respond to. We develop a contract-theoretic model of endogenous clarification in human–AI interaction: strategic under-specification is not driven by AI models becoming better at guessing; it emerges because users strategically economize on prompt specification effort in response to the assistant’s endogenous asking policy when the assistant’s objective diverges from true user welfare. Formalizing this as a Stackelberg game between an assistant (leader) and a population of users with privately known specification costs (followers), we first establish a decision-theoretic benchmark with an exogenous policy. Here, improving raw guessing accuracy alone cannot harm users despite inducing them to specify less—an immediate envelope-theorem benchmark showing that naive “better guessing degrades outcomes” claims conflate technological improvement with policy misalignment.

Closing the loop, we solve for the assistant’s optimal policy under two governance regimes: a *pooling* policy restricted to a single population-wide ask rate, and a *screening* policy offering an incentive-compatible menu conditioned on prompt specification effort. We show that an unbiased (welfare-aligned) assistant achieves full first-best efficiency under screening despite users’ private information—incentive compatibility is free when principal and agent share an objective. Consequently, all screening-regime inefficiency is attributable to misalignment between the assistant’s objective and true user welfare, not to asymmetric information per se. Pooling, by contrast, suffers a structural loss from restricted instruments even when perfectly aligned, collapsing to an extreme bang-bang corner solution. We characterize both equilibria under a tractable linear-risk specification, derive testable comparative statics, provide extensive numerical evidence that the qualitative predictions survive under the exact nonlinear conjunctive specification, establish a 5-level hierarchy of theoretical and computational evidence, and outline a proposed human-subject experimental design.

1 Introduction

A user typing a coding request faces a fundamental economic trade-off: spend cognitive effort articulating edge cases, naming conventions, and constraints, or leave them implicit and trust the assistant to infer them. The assistant, in turn, must decide for each unspecified detail whether to guess or to ask. Neither choice is made in a vacuum: the user’s specification effort depends on what they expect the assistant to do with silence, and the assistant’s ask/guess policy is chosen (explicitly, or as an emergent property of training and product incentives) with some implicit model of how users will respond.

This paper provides a mechanism-design theory of endogenous clarification in AI assistance. Its central premise is that strategic under-specification is not caused by AI getting better at guessing; it is caused by users responding to the assistant’s endogenous clarification policy when the assistant’s

objective differs from theirs. Section 2 begins with a decision-theoretic benchmark where the assistant’s policy is exogenous, proving that improved guessing accuracy alone weakly improves user welfare despite shortening prompts. Section 3 identifies the true moral-hazard channel: unobserved divergence between the user’s belief and the assistant’s true asking policy. Sections 4–7 close the loop by modeling the assistant as a Stackelberg leader anticipating user best responses under pooling and screening instruments. Section 8 translates the equilibrium characterizations into seven falsifiable behavioral predictions (P1–P7). Section 10 evaluates exact nonlinear robustness and discusses institutional extensions.

1.1 Related literature

This work bridges contract theory and the emerging literature on formal models of human–AI interaction.

Contract theory and screening without transfers. Our screening framework builds on classical adverse-selection models (Myerson, 1981; Baron and Myerson, 1982; Mussa and Rosen, 1978; Laffont and Tirole, 1986) that exploit the Spence–Mirrlees single-crossing property (Spence, 1973) to separate heterogeneous agents. However, unlike classical monopoly or regulatory screening where the principal uses monetary transfers to extract information rents, here the assistant screens strictly through an interactive service margin (the ask rate a offered in response to specification effort m). This connects to non-monetary screening and costly verification (Townsend, 1979), where information revelation is governed by operational friction rather than cash transfers.

Costly communication and strategic disclosure. The user’s specification decision formalizes costly disclosure under strategic interpretation, connecting to strategic information transmission (Crawford and Sobel, 1982), voluntary quality disclosure (Grossman, 1981), and rational inattention (Sims, 2003). In our setting, under-specification arises not from an incentive to deceive, but from the user’s rational economizing on specification costs when anticipating automated assistance.

Assistance games and human–AI interaction. Our bi-level interaction connects to the literature on assistance games and cooperative inverse reinforcement learning (Hadfield-Menell et al., 2016; von Stackelberg, 1934), where an autonomous assistant acts to maximize human utility under incomplete information. However, while assistance games formalize the assistant’s uncertainty over the human’s underlying *objective or reward function*, in our setting the economic task objective is known and shared (downstream success value V and error loss L). Instead, the informational asymmetry resides entirely in the user’s private *specification cost* κ . The strategic margin in our model is therefore not reward learning or preference alignment, but rather the contract-theoretic trade-off between prompt specification effort and automated clarification under costly communication. To our knowledge, our model provides the first contract-theoretic microfoundation for this ask-versus-guess margin under asymmetric specification costs.

Clarification and interactive NLP. A parallel empirical literature in natural language processing and information retrieval investigates algorithms for generating and ranking clarifying questions in conversational search and dialog (Rao and Daumé III, 2018; Aliannejadi et al., 2019; Zamani et al., 2020). These works optimize static clarification decisions or train heuristic dialog policies via reinforcement learning against fixed datasets. However, they uniformly treat user inputs as exogenous test instances rather than the output of an optimizing agent who strategically anticipates

the clarification policy. Our framework provides the contract-theoretic microfoundation that this engineering literature lacks, formally endogenizing prompt specification effort as an equilibrium best response to the assistant’s asking policy.

2 Environment and the Decision-Theoretic Benchmark

2.1 Setup

A task has k binary, independent, intent-relevant attributes $\theta = (\theta_1, \dots, \theta_k) \in \{0, 1\}^k$, each drawn i.i.d. uniform and privately known to the user. The user chooses a specification effort level $m \in \{0, \dots, k\}$ and truthfully reveals m attributes.¹ Specifying m attributes costs $c(m)$, with $c(0) = 0$, $c' \geq 0$, $c'' \geq 0$.

For each of the $k - m$ unspecified attributes, the assistant either *asks* (revealing the true value, at a cost to the user) or *guesses*, succeeding independently with probability $g \in (0, 1)$. We model the assistant’s policy at this stage as a uniform ask rate $a \in [0, 1]$ applied to every unspecified attribute, so each is correctly resolved with probability $q(a, g) = a + (1 - a)g$. Task success requires *all* k attributes to be correctly resolved (conjunctive: one wrong parameter or convention typically breaks the deliverable). The user’s payoff is

$$U(m; a, g) = V q(a, g)^{k-m} - c(m) - a(k - m)\kappa,$$

with $V > 0$ the value of success and $\kappa > 0$ the user’s cost of answering one clarifying question. We treat m as continuous on $[0, k]$ for tractability.

2.2 Proposition 1: accuracy alone does not create moral hazard

Proposition 1. *Fix a . Let $U^*(g) = \max_{m \in [0, k]} U(m; a, g)$. Then U^* is weakly increasing in g .*

Proof. By the envelope theorem, at the optimum $m^*(g)$, $dU^*/dg = \partial U / \partial g|_{m=m^*(g)} = V(k - m^*)q^{k-m^*-1}(1 - a) \geq 0$. The envelope theorem lets us ignore dm^*/dg : because m^* is chosen optimally, its first-order response to g has no first-order effect on U^* . $\square$

Assumption 1 (Regularity). *In the relevant range of m , $(k - m) \ln q(a, g) \geq -1$.*

Remark 1 (Interpreting Assumption 1, and Monotonicity Survival). *The condition has no closed-form economic reading on its own; it is a sufficient condition for the single-crossing property used in Corollary 1 and Proposition 2 via Topkis’s monotone comparative statics theorem (Topkis, 1978, 1998; Milgrom and Shannon, 1994), not a claim about primitives directly. The direction is pinned down by the cross-partial itself: for fixed a ,*

$$\frac{\partial^2 U}{\partial m \partial g} = -V(1 - a)q^{k-m-1}[(k - m) \ln q + 1],$$

so U has decreasing differences in (m, g) — the condition Topkis’s theorem (Topkis, 1978, 1998) needs to conclude $m^(g)$ is weakly decreasing — exactly when $(k - m) \ln q + 1 \geq 0$, i.e., when Assumption 1 holds as stated here. If the inequality were reversed, U would exhibit increasing differences and hence $m^*(g)$ would be locally increasing (e.g., at $V=10, k=5, a=0, c(m)=m^2/2$,*

¹We assume truthful revelation conditional on the choice to specify: the strategic margin is *how much* to say, not whether to lie about what is said. A cheap-talk layer on top of this costly-specification layer is a natural extension; see Section 10.

m^* rises from ≈ 0.26 at $g=0.5$ to ≈ 0.67 at $g=0.9$ before turning over and falling as $g \rightarrow 1$). Loosely, since $\ln q \approx -(1-q)$ for q near 1, the condition is close to $(k-m)(1-q) \leq 1$: the expected number of wrong attributes under the current policy is at most about one. A numerical grid evaluation over $k \in [5, 15]$, $q \in [0.70, 0.95]$, and $\kappa \in [0.20, 2.00]$ (726 grid points, quadratic cost $c(m) = \kappa m^2/2$, evaluated at each point's own optimizing m^*) finds that under the benchmark simulation valuation ($V = 100$), the condition holds on 89.9% (653/726) of this parameter space. In lower-valuation regimes where specification effort is lower ($V = 10$, as in the numerical example above), the condition holds on roughly 34% (and 31.7% at $V = 8$); the exact share is sensitive to the scaling of $c(m)$ relative to V . Crucially, across the tested parameter grid (2,552 parameter intervals), a comprehensive finite-difference evaluation confirms that monotonicity of specification ($m^*(g)$ weakly decreasing in g) holds without exception across all 2,162 intervals where Assumption 1 is satisfied (100% survival, zero violations). Outside the sufficient condition ($(k-m)\ln q < -1$, 390 intervals), the cross-partial turns strictly positive and monotonicity breaks in 96.7% of tested intervals (377/390), confirming that the regularity condition is empirically tight.

Corollary 1. Under Assumption 1, $m^*(g)$ is weakly decreasing in g (decreasing differences in (m, g) ; Topkis's theorem).

Proof. Immediate from the cross-partial computed in Remark 1: Assumption 1 is exactly the condition $\partial^2 U / \partial m \partial g \leq 0$ on the relevant range, which is decreasing differences in (m, g) ; Topkis's theorem then gives $m^*(g)$ weakly decreasing. $\square$

So, holding the assistant's policy fixed, users specify less as the assistant gets better at guessing, yet are never worse off for it. We flag explicitly that Proposition 1 is a low bar: it is a direct application of the envelope theorem to a single agent optimizing freely against a technology parameter that enters its payoff with non-negative derivative, and would hold in essentially any model of this form. Its purpose here is not to be a surprising finding but to be the correct benchmark against which any claimed moral hazard should be measured, so that Proposition 2 below can be read as identifying the actual, non-trivial source of harm. It isolates the question this paper is really about: what happens once the policy a is no longer exogenous, but chosen by a strategic assistant that anticipates this response?

3 Sources of Strategic Under-Specification

Proposition 1 assumed the assistant's ask rate is fixed and that users correctly anticipate $q(a, g)$. Both assumptions can fail.

3.1 Cost externality

If a third party (a reviewer, downstream production systems) bears part of the cost of a wrong guess, the user's private (κ, V) understate the social values, and the user under-specifies relative to the social optimum at any fixed g —worsening, by Corollary 1, as g rises. Standard; not our focus.

3.2 Miscalibrated belief updating under silent failure

If wrong guesses are not always detected, and detection probability $d(m)$ is increasing in m , users infer g from experience rather than observing it, and under-specifying users are also less likely to catch their own errors—an upward-biased $\hat{g}$ that reinforces under-specification. This is a dynamic learning story rather than a static equilibrium result; we return to it in the empirical design (prediction P5).

3.3 Belief about the assistant’s policy

Let users be *sophisticated* (correctly infer the assistant’s true ask rate a_A) or *naive* (believe the assistant follows the welfare-optimal rate $a^\dagger \geq a_A$).

Proposition 2. *Under Assumption 1, naive users choose $m_{\text{naive}}^* \leq m_{\text{soph}}^*$ and $U(m_{\text{naive}}^*; a_A, g) < U(m_{\text{soph}}^*; a_A, g)$: naive users under-specify and are strictly worse off than sophisticated users, holding raw guessing accuracy g fixed throughout.*

Proof. Because $q(a, g) = a + (1 - a)g$ is strictly increasing in a for $g \in (0, 1)$, a naive user believing $a^\dagger \geq a_A$ perceives a higher resolution accuracy $q(a^\dagger, g) \geq q(a_A, g)$. Under Assumption 1, Corollary 1 proves that $m^*(g)$ is weakly decreasing in accuracy, so $m_{\text{naive}}^* \leq m_{\text{soph}}^*$. Furthermore, $U(m; a_A, g)$ is strictly concave in m with unique maximizer m_{soph}^* , so any choice $m_{\text{naive}}^* \neq m_{\text{soph}}^*$ delivers strictly lower true utility: $U(m_{\text{naive}}^*; a_A, g) < U(m_{\text{soph}}^*; a_A, g)$. $\square$

This is the source of genuine moral hazard identified so far: not raw accuracy, but an *unobserved gap between the user’s belief about the assistant’s policy and the assistant’s true policy*. Section 5 solves for what that true policy a_A actually is when the assistant is itself a strategic, possibly biased, Stackelberg leader—so Proposition 2 can be instantiated with a derived a_A rather than an assumed one.

4 The Assistant’s Policy as a Stackelberg Game

4.1 General formulation

The assistant (leader) commits ex ante to a policy π before meeting any user. A population of users (followers) has privately known specification-cost type $\kappa \sim F$ on $[\kappa_L, \kappa_H]$, with cost $c(m; \kappa) = \kappa m^2/2$, and best-responds $m^*(\kappa; \pi) = \arg \max_m U(m; \pi, \kappa)$. The leader’s own objective Π need not equal U : we write

$$\Pi_\kappa(m, a) = \mu_A U(m, a; \kappa) - (\lambda_A - c_Q) a (k - m),$$

where $\mu_A \in (0, 1]$ discounts the leader’s weight on true downstream user welfare, c_Q is the user’s true cognitive cost of answering one clarifying question, and λ_A is the leader’s own perceived or institutional cost per clarifying question. In commercial AI systems, misalignment along these margins arises from concrete engineering and economic forces:

1. *Asking friction bias* ($\lambda_A > c_Q$): Reinforcement learning from human feedback (RLHF) and preference optimization algorithms (e.g., DPO) commonly evaluate conversation turns in isolation. Reward models systematically penalize “evasive” clarifying questions in favor of immediate code generation because users in rating pools favor direct answers. Furthermore, multi-turn interactions increase token costs, context-window expansion, and inference latency, creating provider-side pressure to suppress clarification turns.
2. *Downstream welfare discounting* ($\mu_A < 1$): Turn-level user satisfaction signals (such as prompt acceptance or thumbs-up ratings) capture immediate session fluency. In contrast, the true economic loss L from broken conventions, subtle edge-case errors, or security defects is typically discovered hours or days later in downstream execution, CI/CD testing, or code review—leading training objectives tuned to immediate session retention to discount downstream correctness.

The leader is *unbiased* iff $\mu_A = 1$ and $\lambda_A = c_Q$, in which case $\Pi_\kappa \equiv U(\cdot; \kappa)$ exactly. The leader solves $\pi^* = \arg \max_\pi \mathbb{E}_{\kappa \sim F} [\Pi_\kappa(m^*(\kappa; \pi), \pi)]$, a bilevel program.

Remark 2 (Existence). *For Model I below ($\pi = a \in [0, 1]$, a compact scalar policy), $m^*(\kappa; a)$ is continuous in a (strict concavity of U in m plus the implicit function theorem), so $\mathbb{E}_\kappa[\Pi_\kappa]$ is continuous in a and a maximizer exists on $[0, 1]$ by Weierstrass’s theorem. For Model II below (a finite menu over two types), the feasible set is a compact subset of $\mathbb{R}^4$ cut out by linear IC/IR constraints and a continuous objective, so a maximizer again exists by Weierstrass’s theorem; no ironing or ex-ante randomization is needed with only two types.*

For tractability in what follows we replace the exact conjunctive success probability q^{k-m} with its first-order (small-risk) approximation around $q \rightarrow 1$: expected loss $\approx L(1 - q)(k - m)$, where L denotes the economic loss from a single unresolved-and-incorrect attribute. With $\Lambda \equiv L(1 - g)$, the user’s payoff becomes

$$U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2}m^2 - a c_Q (k - m).$$

Throughout Sections 4.2–6, all closed-form characterizations are stated for the interior domain $\kappa \geq \Lambda/k$, ensuring that the unconstrained optimal specification $m^*(\kappa; a) = \frac{\Lambda(1-a)+ac_Q}{\kappa}$ lies strictly in $[0, k]$ for all $a \in [0, 1]$ under the maintained condition $\Lambda > c_Q$. For low-cost types ($\kappa < \Lambda/k$), best-response effort saturates at the physical attribute capacity $m^* = k$, resolving all attributes upstream. As demonstrated in Section 10 and verified in our computational test suite, the qualitative regime structure and corner-solution properties survive boundary clipping identically.

4.2 The first-best benchmark

Before analyzing either regime, characterize the fully-informed, fully-aligned optimum: $(m^{FB}(\kappa), a^{FB}(\kappa)) = \arg \max_{m,a} U(m, a; \kappa)$.

Proposition 3 (First-best is a threshold rule in κ). *On the interior domain $\kappa \geq \Lambda/k$, U is linear in a for fixed m , so $a^{FB}(\kappa) \in \{0, 1\}$. Define $\kappa^* = (\Lambda + c_Q)/(2k)$. If $\Lambda > c_Q$, then $a^{FB}(\kappa) = 1$ (always ask) for $\kappa > \kappa^*$ and $a^{FB}(\kappa) = 0$ (never ask, guess) for $\kappa < \kappa^*$.*

Proof. At $a = 1$: $m^*(\kappa; 1) = c_Q/\kappa$ and $U(m^*, 1; \kappa) = V - c_Q k + c_Q^2/(2\kappa)$. At $a = 0$: $m^*(\kappa; 0) = \Lambda/\kappa$ and $U(m^*, 0; \kappa) = V - \Lambda k + \Lambda^2/(2\kappa)$. Subtracting,

$$U(a=1) - U(a=0) = [\Lambda - c_Q] \left(k - \frac{\Lambda + c_Q}{2\kappa} \right),$$

which is positive iff $\Lambda > c_Q$ and $\kappa > (\Lambda + c_Q)/(2k) = \kappa^*$. $\square$

Remark 3 (Boundary types $\kappa < \Lambda/k$). *For low-cost types with $\kappa < \Lambda/k$, unconstrained first-best effort exceeds k , so specification saturates at $m^{FB} = k$. At $m = k$, all attributes are resolved upstream ($k - m = 0$), rendering the asking policy payoff-irrelevant ($U(k, a; \kappa) = V - \kappa k^2/2$).*

This is a clean, closed-form, type-dependent first best: high-cost (terse) users, who cannot cheaply self-provide detail, should always be asked; low-cost (verbose) users, who can cheaply write enough that the small residual is safe to guess, should never be asked. This threshold rule completely characterizes the efficient asking policy as a function of the user’s marginal cost of detail relative to downstream error loss.

5 Model I: A Pooling (Scalar) Policy

The leader is restricted to a single population-wide ask rate $a \in [0, 1]$ —it cannot condition on anything user-specific. This is the natural benchmark for a policy that is fixed at training time and cannot observe individual users.

5.1 User best response

The FOC $\partial U / \partial m = \Lambda(1 - a) - \kappa m + a c_Q = 0$ gives, with $\beta(\kappa) = \Lambda / \kappa$ and $\gamma(\kappa) = (\Lambda - c_Q) / \kappa$,

$$m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa), \quad \frac{\partial m^*}{\partial a} = -\gamma(\kappa) = \frac{c_Q - \Lambda}{\kappa}.$$

We assume interior specification levels $m^* \in (0, k)$; for low-cost types where unconstrained $\beta(\kappa) \geq k$, specification effort saturates at $m^* = k$. The closed-form population moments $\bar{R}_0$ and $\bar{\gamma}$ below are exact when best responses remain interior, and serve as tractable linear approximations when boundary clipping binds; as verified in Section 10 and our computational robustness suite, the qualitative regime structure and corner-solution properties are strictly preserved under boundary clipping and exact payoffs alike.

Remark 4 (Non-monotone effect of ask rate on specification). *Unlike Corollary 1, the sign of $\partial m^* / \partial a$ here does not depend on type: it is positive for every type iff $c_Q > \Lambda$, i.e., iff answering a question costs users more than the expected loss it avoids. When that holds, a higher population ask rate induces users to specify more, not less—they pre-empt costly clarifying questions by writing the answer down themselves. This is a genuinely new force relative to Section 2, where a only entered through accuracy, and is a distinct, directly testable prediction (P1' below).*

5.2 The leader's problem

Let $R(a, \kappa) = k - m^*(\kappa; a)$ denote the residual unspecified attributes. In the pooling regime, the assistant commits ex ante to an operational ask rate $a \in [0, 1]$ applied population-wide, without observing or directly contracting on individual specification effort.

Evaluating the general leader objective from Section 4.2 at the follower's best response $m^*(\kappa; a) = \beta(\kappa) - a \gamma(\kappa)$:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A U(m^*(\kappa; a), a; \kappa) - (\lambda_A - c_Q) a (k - m^*(\kappa; a)).$$

Substituting $U(m^*(\kappa; a), a; \kappa) = V - \Lambda(1 - a)(k - m^*) - \frac{\kappa}{2}(m^*)^2 - a c_Q (k - m^*)$, using $k - m^*(\kappa; a) = R_0(\kappa) + a \gamma(\kappa)$, and defining the effective resolution stake $s \equiv \Lambda - c_Q$ and the assistant's excess asking friction $b \equiv \lambda_A - c_Q$, expanding in powers of a yields:

$$\Pi_\kappa(m^*(\kappa; a), a) = \mu_A \left(V - \Lambda k + \frac{\Lambda^2}{2\kappa} \right) + (\mu_A s - b) R_0(\kappa) a + \frac{(\mu_A s - 2b) \gamma(\kappa)}{2} a^2.$$

Averaging over $\kappa \sim F$ requires no moments beyond the population sufficient statistics already defined, $\bar{R}_0 = k - \mathbb{E}[\beta]$ and $\bar{\gamma} = \mathbb{E}[\gamma] = s \mathbb{E}[1/\kappa]$:

$$\Pi(a) = \text{const} + \bar{L} a + \bar{Q} a^2,$$

where $\text{const} = \mu_A (V - \Lambda k) + \frac{\mu_A \Lambda^2}{2} \mathbb{E}[1/\kappa]$, and

$$\bar{L} \equiv (\mu_A s - b) \bar{R}_0, \quad \bar{Q} \equiv \frac{(\mu_A s - 2b) \bar{\gamma}}{2}.$$

Proposition 4 (Stackelberg-optimal pooling rate). *Provided $\bar{Q} < 0$ (the second-order condition for strict concavity, equivalent to $2(\lambda_A - c_Q) > \mu_A(\Lambda - c_Q)$), the leader's unconstrained stationary point is*

$$a^{SE} = -\frac{\bar{L}}{2\bar{Q}} = -\frac{(\mu_A s - b) \bar{R}_0}{(\mu_A s - 2b) \bar{\gamma}}.$$

When this stationary point lies in $[0, 1]$, it is the unique global maximum of $\Pi(a)$ on $[0, 1]$.

Corollary 2 (Bias shifts the pooling rate on the interior branch). *Differentiating the closed form in Proposition 4 directly with respect to λ_A (holding primitives $\mu_A, \Lambda, c_Q, \bar{R}_0, \bar{\gamma}$ fixed) gives, on the interior branch $\bar{Q} < 0$,*

$$\frac{\partial a^{SE}}{\partial \lambda_A} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A s}{(\mu_A s - 2b)^2} = -\frac{\bar{R}_0}{\bar{\gamma}} \frac{\mu_A(\Lambda - c_Q)}{[\mu_A(\Lambda - c_Q) - 2(\lambda_A - c_Q)]^2},$$

which is strictly positive whenever $\bar{R}_0 < 0$ (the under-specification regime). Hence, a^{SE} is strictly increasing in perceived friction λ_A throughout the interior branch. This comparative static holds on the interior branch ($\bar{Q} < 0$); for the global optimum in the corner regime, see Corollary 4.

Corollary 3 (Non-identification of the two bias channels in the interior regime). *On the interior branch ($\bar{Q} < 0$), defining the normalized excess friction per unit of altruism $\tilde{\rho} \equiv \frac{\lambda_A - c_Q}{\mu_A} = \frac{b}{\mu_A}$, the optimal pooling rate simplifies to*

$$a^{SE} = -\frac{\bar{R}_0}{\bar{\gamma}} \left[\frac{s - \tilde{\rho}}{s - 2\tilde{\rho}} \right],$$

so a^{SE} depends on the two bias parameters (μ_A, λ_A) strictly through the single scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$ (holding primitives s and $\bar{R}_0/\bar{\gamma}$ fixed). Consequently,

$$\frac{\partial a^{SE}/\partial \mu_A}{\partial a^{SE}/\partial \lambda_A} = -\frac{\lambda_A - c_Q}{\mu_A} = -\tilde{\rho},$$

and any two parameter pairs on the same ray $(\lambda_A - c_Q)/\mu_A = \text{const}$ produce the identical pooling ask rate. A leader that discounts downstream welfare ($\mu_A < 1$, $\lambda_A = c_Q \implies \tilde{\rho} = 0$) cannot be distinguished, from a^{SE} alone, from any leader along the corresponding ray of excess friction.

Remark 5 (Scope of Corollary 3). *The derivation above is stated on the interior branch $\bar{Q} < 0$. By Corollary 5 below, the unbiased pooling optimum is always a corner ($\bar{Q} \geq 0$), so the interior stationary point describes behavior when friction bias is sufficiently large ($2b > \mu_A s$).*

Corollary 4 (Ray-invariance extends to the corner regime). *On the corner branch ($\bar{Q} \geq 0$, or whenever the unconstrained stationary point lies outside $[0, 1]$), the optimal policy is chosen by the sign of the corner difference $\Pi(1) - \Pi(0) = \bar{L} + \bar{Q}$:*

$$\Pi(1) - \Pi(0) = \mu_A \left[s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) - \tilde{\rho}(\bar{R}_0 + \bar{\gamma}) \right].$$

Since $\mu_A > 0$, μ_A factors out completely, and the sign depends solely on the scalar ratio $\tilde{\rho} = (\lambda_A - c_Q)/\mu_A$. The critical threshold ratio is

$$\tilde{\rho}^* \equiv \frac{s(\bar{R}_0 + \bar{\gamma}/2)}{\bar{R}_0 + \bar{\gamma}},$$

with $a^{SE} = 1$ for $\tilde{\rho} < \tilde{\rho}^$ and $a^{SE} = 0$ for $\tilde{\rho} > \tilde{\rho}^*$. Ray-invariance in the excess friction ratio $\tilde{\rho}$ therefore holds unconditionally across the entire parameter space, interior and corner regimes alike. A numerical sweep of 11 rays across 10 values of $\mu_A \in [0.1, 1.0]$ (110 evaluations) confirms this analytical threshold with zero boundary-crossing violations.*

Corollary 5 (Pooling is a corner solution in the unbiased case). *In the unbiased case ($\mu_A = 1, \lambda_A = c_Q \implies b = 0$), the quadratic coefficient is:*

$$\bar{Q} = \frac{s^2 \bar{\gamma}}{2} = \frac{(\Lambda - c_Q)^2 \mathbb{E}[1/\kappa]}{2} \geq 0 \quad \text{always.}$$

The second-order condition for an interior maximum therefore fails: $\Pi(a)$ is weakly convex on $[0, 1]$, its stationary point is a local minimum, and the true optimum is always a corner, $a^{SE} \in \{0, 1\}$, not an interior compromise. Comparing the two corners directly:

$$\Pi(1) - \Pi(0) = \bar{L} + \bar{Q} = s \left(\bar{R}_0 + \frac{\bar{\gamma}}{2} \right) = (\Lambda - c_Q) \left[k - \frac{\Lambda + c_Q}{2} \mathbb{E} \left[\frac{1}{\kappa} \right] \right],$$

so the unbiased pooling optimum is $a^{SE} = 1$ (always ask) when $k \geq \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$ and $a^{SE} = 0$ (never ask) when $k < \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$. This threshold matches the population-averaged first-best preference $\mathbb{E}[U(1) - U(0)]$ from Proposition 3 identically.

This last point matters for the comparison in Section 7: Model I has two independent, additive sources of loss relative to first best—the pooling restriction, and bias.

6 Model II: A Screening (Menu) Policy

Now let the leader condition its ask rate on the user's declared specification level (equivalently, by the revelation principle, offer a menu of bundles that users self-select into). For tractability and clean analytical exposition, we restrict attention to two types $\kappa_L < \kappa_H$ with population shares f_L, f_H (the continuum-type extension using the standard Mirrlees technique is discussed in Section 10). Let the leader choose bundles $(m_L, a_L), (m_H, a_H)$ subject to

$$\text{IC}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq U(m_{\kappa'}, a_{\kappa'}; \kappa), \quad \text{IR}_\kappa : U(m_\kappa, a_\kappa; \kappa) \geq \underline{U},$$

to maximize $f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H)$.

6.1 Single-crossing

The marginal rate of substitution between m and a along an indifference curve is $\text{MRS}(m, a; \kappa) = -(\partial U / \partial m) / (\partial U / \partial a)$. From user utility $U(m, a; \kappa) = V - \Lambda(1 - a)(k - m) - \frac{\kappa}{2} m^2 - ac_Q(k - m)$, we have $\partial U / \partial m = \Lambda(1 - a) + ac_Q - \kappa m$ and $\partial U / \partial a = s(k - m)$, where $s \equiv \Lambda - c_Q > 0$ is the net social gain from asking. Hence,

$$\text{MRS}(m, a; \kappa) = \frac{\kappa m - [\Lambda - as]}{s(k - m)}, \quad \frac{\partial \text{MRS}}{\partial \kappa} = \frac{m}{s(k - m)} > 0,$$

which holds strictly for all interior specification levels $0 < m < k$ under the maintained condition $s > 0$. Higher- κ types therefore have strictly steeper indifference curves in (m, a) -space: they require more compensating ask-service per unit of specification effort given up. This satisfies the standard Spence–Mirrlees single-crossing condition (Spence, 1973) and underwrites the screening analysis below.

6.2 No distortion without bias

Proposition 5 (Efficiency survives private information when the leader is unbiased). *If $\mu_A = 1$ and $\lambda_A = c_Q$ (so $\Pi_\kappa \equiv U(\cdot; \kappa)$ for every κ), then implementing each type's own first-best bundle, $(m_\kappa, a_\kappa) = (m^{FB}(\kappa), a^{FB}(\kappa))$ from Proposition 3, is incentive compatible, and no further mechanism design can do better for the leader.*

Proof. $(m^{FB}(\kappa), a^{FB}(\kappa))$ maximizes $U(\cdot; \kappa)$ by definition, so for any $\kappa' \neq \kappa$, $U(m^{FB}(\kappa), a^{FB}(\kappa); \kappa) \geq U(m^{FB}(\kappa'), a^{FB}(\kappa'); \kappa)$ trivially: no bundle, including another type's, can give type κ more utility than its own maximizer. IC holds with no distortion needed. Since $\Pi_\kappa \equiv U(\cdot; \kappa)$, the leader's payoff from serving each true type is exactly maximized at that type's own first best, so this menu is also leader-optimal. $\square$

This is the paper's sharpest result: unlike classical regulation (Baron–Myerson, Laffont–Tirole), where the principal pays a literal transfer and therefore has an independent reason to dislike conceding rent, here the leader's payoff can coincide exactly with user welfare, and when it does, incentive compatibility is *free*—private information costs nothing. Distortion in this setting is therefore a pure signature of misalignment, not of asymmetric information per se.

6.3 Distortion under bias

When $\Pi_\kappa \neq U(\cdot; \kappa)$ (bias present), the leader's preferred bundle per type, $(m_\kappa^B, a_\kappa^B) = \arg \max_{m,a} \Pi_\kappa(m, a)$ (computable in closed form by the same method as Proposition 3, with Λ, c_Q replaced by their μ_A, λ_A -weighted counterparts), no longer automatically satisfies IC, because Π_κ and the user's true $U(\cdot; \kappa)$ now rank bundles differently.

Proposition 6 (Downward distortion of the high-cost type's service). *Assume $\Lambda > c_Q$, $\kappa_L < \kappa^{*B} < \kappa_H$, and under-asking bias ($\lambda_A > \mu_A c_Q$, so $c_Q^{eff} \equiv c_Q + \frac{\lambda_A - c_Q}{\mu_A} > c_Q$). Then:*

1. **Incentive violation of unconstrained menu:** *The unconstrained biased menu $(m_L^B, a_L^B) = (\Lambda/\kappa_L, 0)$ and $(m_H^B, a_H^B) = (c_Q^{eff}/\kappa_H, 1)$ strictly violates IC_L whenever the type heterogeneity spread $\kappa_H - \kappa_L$ is sufficiently large.*
2. **Downward distortion:** *Under the optimal incentive-compatible menu, the low-cost type receives no distortion ($a_L^{SB} = a_L^B = 0$), while the high-cost type's ask rate is strictly distorted downward:*

$$a_H^{SB} < a_H^B \leq a_H^{FB} = 1.$$

3. **Active constraint regime:** *In the moderate-bias regime, IC_L binds alone while IR_H , IC_H , and IR_L remain strictly slack. The reverse constraint IC_H becomes co-active if and only if asking friction is extreme ($\lambda_A \geq \lambda_A^{pool}$), where the screening menu collapses into a pooling contract.*

Proof. We establish the result in five analytical steps:

Step 1: Unconstrained biased optima. The leader objective per type is $\Pi_\kappa(m, a) = \mu_A[V - \Lambda(1-a)(k-m) - \frac{\kappa}{2}m^2 - ac_Q^{eff}(k-m)]$, where $c_Q^{eff} \equiv c_Q + (\lambda_A - c_Q)/\mu_A > c_Q$ under under-asking bias. Maximizing unconstrained with respect to m yields $m^B(\kappa; a) = [\Lambda(1-a) + ac_Q^{eff}]/\kappa$. Because Π_κ is linear in a for fixed m with slope $\mu_A(\Lambda - c_Q^{eff})(k-m)$, the biased first-best ask rate is bang-bang around the threshold $\kappa^{*B} = (\Lambda + c_Q^{eff})/(2k)$. For $\kappa_L < \kappa^{*B} < \kappa_H$, the unconstrained menu is $(m_L^B, a_L^B) = (\Lambda/\kappa_L, 0)$ and $(m_H^B, a_H^B) = (c_Q^{eff}/\kappa_H, 1)$.

Step 2: Upward temptation to mimic (IC_L violation). Evaluating type κ_L 's payoff from accepting the high type's bundle $(m_H^B, 1)$ versus its own bundle $(m_L^B, 0)$:

$$U(m_H^B, 1; \kappa_L) - U(m_H^B, 1; \kappa_H) = \frac{\kappa_H - \kappa_L}{2} (m_H^B)^2 > 0.$$

Because type κ_L can provide the high type's specification level $m_H^B = c_Q^{eff}/\kappa_H$ at strictly lower cognitive cost ($\kappa_L < \kappa_H$), its utility from mimicking type H exceeds type H 's own utility by the informational rent $\frac{\kappa_H - \kappa_L}{2} (m_H^B)^2$. When $\kappa_H - \kappa_L$ is sufficiently wide, $U(m_H^B, 1; \kappa_L) > U(m_L^B, 0; \kappa_L)$, violating IC_L.

Step 3: Slackness of participation (IR_H). In classical adverse selection with monetary transfers (Baron–Myerson, Laffont–Tirole), the principal's payoff is linearly reduced by agent rent, driving the inefficient agent's IR to bind at reservation $\underline{U}$. Here, the principal's payoff Π_{κ_H} directly incorporates $\mu_A U(\cdot; \kappa_H)$. Depressing U_H directly penalizes the leader's own objective. Thus, the leader adjusts the bundle (m_H, a_H) only as far as necessary to restore IC_L, leaving IR_H strictly slack: $U(m_H^{SB}, a_H^{SB}; \kappa_H) > \underline{U}$.

Step 4: Direction of distortion along the IC_L boundary. Form the Lagrangian of the relaxed problem where IC_L binds with multiplier $\mu_{IC_L} > 0$:

$$\mathcal{L} = f_L \Pi_{\kappa_L}(m_L, a_L) + f_H \Pi_{\kappa_H}(m_H, a_H) + \mu_{IC_L} [U(m_L, a_L; \kappa_L) - U(m_H, a_H; \kappa_L)].$$

By single-crossing ($\partial \text{MRS} / \partial \kappa > 0$), no downward constraint binds at the optimum ($a_L^{SB} = 0, m_L^{SB} = m_L^B$). Differentiating $\mathcal{L}$ with respect to a_H :

$$\frac{\partial \mathcal{L}}{\partial a_H} = f_H \frac{\partial \Pi_{\kappa_H}}{\partial a_H} - \mu_{IC_L} \frac{\partial U(m_H, a_H; \kappa_L)}{\partial a_H} = 0.$$

From user utility, $\partial U(m_H, a_H; \kappa_L) / \partial a_H = s(k - m_H)$, where $s \equiv \Lambda - c_Q > 0$. Therefore, at the constrained optimum:

$$\frac{\partial \Pi_{\kappa_H}}{\partial a_H} = \frac{\mu_{IC_L}}{f_H} s(k - m_H).$$

Because $\mu_{IC_L} > 0$ (the constraint binds) and $s(k - m_H) > 0$ for any interior $m_H < k$, the right-hand side is strictly positive. Because Π_{κ_H} is strictly concave in a near the unconstrained optimum (where $\partial \Pi / \partial a = 0$ at a_H^B), a strictly positive marginal value $\partial \Pi_{\kappa_H} / \partial a_H > 0$ requires the constrained optimum a_H^{SB} to lie strictly to the left of the unconstrained peak:

$$a_H^{SB} < a_H^B \leq 1.$$

Hence, the high-cost type's asking rate is strictly distorted downward.

Step 5: Boundary of IC_H co-activity. As bias λ_A increases, maintaining IC_L forces a_H^{SB} progressively lower. When a_H^{SB} drops toward $a_L^{SB} = 0$, the high-cost type receives negligible clarification assistance. Eventually, the reverse constraint IC_H ($U(m_H, a_H; \kappa_H) \geq U(m_L, a_L; \kappa_H)$) binds when $U(m_H^{SB}, a_H^{SB}; \kappa_H) = U(m_L^B, 0; \kappa_H)$. Beyond this threshold $\lambda_A \geq \lambda_A^{\text{pool}}$, separating the types is no longer feasible, and the menu collapses into the pooling policy of Model I. $\square$

Remark 6 (Active constraint structure without monetary transfers). *In classical transfer-based screening (e.g., Baron–Myerson), IC_L and IR_H typically bind while IC_H and IR_L remain slack. In our setting, solving the full four-variable constrained problem shows that IC_L binds alone, with IR_H , IC_H , and IR_L remaining strictly slack at representative under-asking-bias parameters. This structural divergence arises because the leader’s payoff Π_{κ_H} directly internalizes follower welfare $\mu_A U(\cdot; \kappa_H)$, eliminating the classical principal’s motive to compress agent rent down to the reservation utility. We note that the exact active constraint configuration is parameter-region-specific: under sufficiently extreme asking friction ($\lambda_A \gg c_Q$) or severe discounting ($\mu_A \ll 1$), IC_H can bind alongside IC_L . The robust qualitative insight is that screening under misalignment strictly distorts $a_H^{SB} < a_H^B$, while the active constraint configuration reflects the non-monetary, partially aligned nature of the objective.*

Remark 7. *The direction in Proposition 6 is specific to the under-asking bias case ($\lambda_A > \mu_{ACQ}$), which we take to be the empirically relevant one (Section 3.3’s motivating story: turn-level satisfaction signals penalize extra clarifying questions more than users actually mind answering them). Under the opposite bias, the direction of distortion can flip; we do not claim a sign that holds for all Π_κ .*

7 Comparing the Two Regimes

Corollary 6 (Menus weakly dominate pooling). *The leader’s optimal expected payoff under Model II is weakly greater than under Model I: any pooling policy a is a feasible (degenerate) menu with $a_L = a_H = a$, trivially satisfying IC , so the unconstrained menu optimum is at least as good.*

The more interesting comparison is *where the two regimes’ losses come from*, which Propositions 4–6 let us decompose cleanly:

- **Model I (pooling)** loses welfare from two distinct sources: (i) the pooling restriction itself, present even when $\mu_A=1, \lambda_A=c_Q$, where it manifests as a corner solution that fully serves one type and fully fails the other (Corollary 5), and (ii) bias, which shifts a^{SE} (Corollary 2) and can additionally shift pooling between corner and interior regimes. The two sources interact non-linearly across the parameter space.
- **Model II (screening)** loses welfare from bias *only* (Proposition 5): with no bias, screening recovers full efficiency despite private information. With bias, the loss is the rent-extraction-driven downward distortion of Proposition 6, which is a genuinely different mechanism from Model I’s pooling loss (it doesn’t arise from being unable to condition on anything—it arises from not being able to trust what is conditioned on).

7.1 Sensitivity of the Unbiased Pooling Corner

While Corollary 5’s qualitative finding—that unbiased pooling is always a corner solution rather than an interior compromise—is structural and universal, the specific corner selected depends on whether the task size k exceeds the population threshold $k^* \equiv \frac{\Lambda + c_Q}{2} \mathbb{E}[1/\kappa]$. Table 1 illustrates this boundary across varying task sizes k and baseline guessing accuracies g , holding $L = 10, c_Q = 2$, and $\kappa \sim \text{Uniform}[0.2, 0.6]$ (where $\mathbb{E}[1/\kappa] \approx 2.747$).

Remark 8 (Compounding, testable). *Because Model II’s distortion is triggered by, and scales with, the same bias parameters (μ_A, λ_A vs. c_Q) that drive Model I’s shift in a^{SE} , reducing measured*

g	$\Lambda = L(1 - g)$	Threshold k^*	Task Size k	$\Pi(1) - \Pi(0)$	Optimal Corner a^{SE}
0.30	7.0	12.36	5	-36.80	0 (never ask)
0.30	7.0	12.36	10	-11.80	0 (never ask)
0.30	7.0	12.36	15	+13.20	1 (always ask)
0.50	5.0	9.61	5	-13.84	0 (never ask)
0.50	5.0	9.61	9	-1.84	0 (never ask)
0.50	5.0	9.61	10 (Benchmark)	+1.16	1 (always ask)
0.50	5.0	9.61	12	+7.16	1 (always ask)
0.50	5.0	9.61	15	+16.16	1 (always ask)
0.70	3.0	6.87	5	-1.87	0 (never ask)
0.70	3.0	6.87	8	+1.13	1 (always ask)
0.70	3.0	6.87	10	+3.13	1 (always ask)

Table 1: Sensitivity of the unbiased pooling corner choice $\Pi(1) - \Pi(0) = (\Lambda - c_Q)[k - k^*]$ across task complexity k and guessing capability g , for $L = 10, c_Q = 2, \kappa \sim \text{Uniform}[0.2, 0.6]$. At the paper’s canonical benchmark ($g = 0.5, k = 10$), asking yields a modest advantage of +1.16, which flips to never-asking at $k \leq 9$.

assistant bias should improve outcomes for high-cost (terse) users under both regimes, but the welfare gap between menu-conditioned and pooled policies should track population heterogeneity in κ (the spread $\kappa_H - \kappa_L$ and how far F sits on either side of κ^) almost independently of bias. This gives two separable, testable quantities from the same field data: a bias-sensitive term and a heterogeneity-sensitive term.*

8 Testable Predictions

- P1** (Prop. 1, Cor. 1) Holding the assistant’s policy fixed, higher guessing accuracy should be associated with shorter specifications but higher or equal success rates—never both shorter specs and lower success.
- P1’** (Model I remark) Holding accuracy fixed, the effect of a higher *ask rate* on specification length should flip sign depending on whether $c_Q \gtrless \Lambda = L(1 - g)$: where clarifying questions are cheap relative to the risk they resolve, users should specify less as ask rate rises (substitution); where questions are relatively costly to answer, users should specify *more* (pre-emption). This is a within-model prediction distinct from P1’s accuracy channel.
- P2** (Prop. 2) The correlation between turn-level satisfaction and eventual task success should weaken, or invert, specifically for under-asking assistants, and can now be benchmarked against the derived a^{SE} of Proposition 4 rather than an assumed ask rate.
- P3** Making ask-rate statistics visible to users should shift specification length toward the sophisticated best response of Section 3.3.
- P4** (Prop. 3) Ask rates should rise with a user’s revealed difficulty specifying detail (terseness), sharply around a threshold κ^* , rather than responding only to model confidence.
- P5** Users with lower self-reported error-catching rates should show declining specification effort over a session (Section 3.2’s dynamic channel).

- P6** (Cor. 3) Under a pooling policy, terse users should be under-served and verbose users over-served relative to *each type’s own* first best, even for a demonstrably unbiased assistant—a gap attributable purely to instrument restriction, distinguishable in data from bias by holding λ_A fixed (e.g., via an A/B test on friction penalties in training) and varying only whether the policy can condition on declared specification.
- P7** (Prop. 6) Even where an assistant *can* condition on declared specification, terse users should still be asked less than the assistant’s own (possibly already-biased) per-type preference would predict, specifically because of the incentive-compatibility constraint—a distortion that should shrink as the population’s cost heterogeneity ($\kappa_H - \kappa_L$) shrinks, holding bias fixed (Remark on compounding).

9 Empirical Strategy and Hierarchy of Evidence

To ensure complete methodological transparency, our theoretical claims and computational evaluations are organized into five distinct levels of evidence:

1. **Level 1 — Analytical Theorems:** Core equilibrium characterizations (First-Best bang-bang threshold in Proposition 3, pooling closed form in Proposition 4, unbiased corner pooling in Corollary 5, single-crossing in Section 6, full efficiency in Proposition 5, and downward screening distortion in Proposition 6) proven formally from primitives under the linear-risk specification.
2. **Level 2 — Exact Nonlinear Robustness:** Numerical constrained optimizations and grid sweeps demonstrating that the qualitative equilibrium structure (corner pooling, downward distortion, and extreme-bias regimes) survives when replacing the linear-risk approximation with the exact conjunctive success function $q(a, g)^{k-m}$ (Section 10).
3. **Level 3 — Numerical Primitive Verification:** Direct computational checks in our accompanying test suite that verify analytical closed forms against fine-grid argmax searches, check second-order conditions, match finite-difference comparative statics to analytical derivatives within 1.8%, and audit active Kuhn–Tucker constraint sets directly from numerical Lagrange multipliers.
4. **Level 4 — Falsifiable Behavioral Predictions:** Seven distinct, directional empirical predictions (P1–P7 in Section 8) that translate theoretical mechanisms into observable margins for human-subject and institutional experimentation.
5. **Level 5 — Agent-Based Simulation Infrastructure:** An extensible software pipeline in the accompanying codebase that instantiates simulated user and assistant agents interacting under derived policies, with deterministic dry-run stubs for reproducible execution. We emphasize that this layer provides computational demonstration of the interaction protocol, not behavioral evidence of human or LLM decision-making.

Proposed human-subject study design. Testing predictions P1’, P6, and P7 requires moving beyond observational log analysis where assistant bias λ_A and user cost κ are confounded. We propose a between-subjects randomized controlled trial on a population of software developers performing standardized coding tasks:

- *Treatments:* Users are randomly assigned to assistants operating under: (A) Low asking policy ($a = 0.1$), (B) High asking policy ($a = 0.8$), (C) Menu-conditioned screening policy (where prompt length endogenously selects the clarification rate), and (D) Unbiased benchmark.

- *Measurements:* For each task, we log upstream prompt length, number of specified edge cases/invariants, turn-level clarification counts, downstream test suite pass rate, and post-session subjective satisfaction.
- *Identification:* By exogenously varying a and measuring whether terse users expand prompt effort when questions are expensive ($c_Q > \Lambda$, testing P1'), this design isolates the causal effect of clarification policies on human specification economizing.

10 Discussion and Limitations

The linear-risk approximation to the conjunctive success function is the main tractability assumption carried through Sections 4–7. To verify that this simplification does not drive our core theoretical findings, we evaluate both the unbiased pooling optimum and the biased screening menu directly under the exact conjunctive payoff $q(a, g)^{k-m}$ (independently of the linear-risk closed forms, via direct numerical grid search and constrained optimization). Across our tested parameter grid ($k \in [3, 20]$ and $g \in [0.50, 0.95]$), we find strong numerical evidence that the qualitative results survive under the exact nonlinear specification: the corner-solution property of Corollary 5 survives in 100% of tested configurations ($a_{\text{exact}}^{SE} \in \{0, 1\}$); downward distortion ($a_H^{SB} < a_H^B$) persists across the grid, with exceptions concentrated at small k where specification effort saturates at $m = k$; and participation constraints remain strictly slack. Furthermore, a comprehensive parameter sweep of the full four-variable constrained menu problem under the exact conjunctive payoff across $\lambda_A \in [c_Q, 10c_Q]$ and $\mu_A \in [0.1, 1.0]$ (100 grid points) confirms that IC_H binds alongside IC_L under extreme bias, reproducing the exact three-regime structure (no binding constraint, IC_L alone, $IC_L + IC_H$ together) identified under the linear-risk approximation, with only the boundary between moderate- and extreme-bias regimes shifting slightly (from $\lambda_A \approx 6.0$ to $\lambda_A \approx 8.0$ at $\mu_A = 1.0$). The extreme-bias reversal is therefore an intrinsic structural feature of the screening contract space, not an artifact of the linear-risk approximation.

Regarding the regularity condition (Assumption 1), our computational audit confirms that monotonicity of prompt specification ($m^*(g)$ weakly decreasing in g) holds across 100% of tested intervals satisfying $(k - m) \ln q \geq -1$. While monotonicity breaks in 96.7% of intervals where the cross-partial turns positive, we frame the survival of monotonicity outside strict submodularity boundaries in lower-stakes regions as an intriguing *numerical robustness finding*: Topkis submodularity is a sufficient condition, but empirical monotonicity can survive locally over wider parameter domains.

A stylized modeling assumption is attribute symmetry (k binary attributes with identical loss L and guessing probability g). In practical software engineering, attributes exhibit severe loss heterogeneity: omitting an architectural invariant or security convention causes catastrophic failure ($L_i \gg 0$), while omitting a variable-naming convention carries negligible cost ($L_j \approx 0$). In an extension with attribute-specific losses L_i , the user's objective becomes $U = V \prod_i q_i - c(m) - \sum_i a_i c_Q$. Ordering attributes by descending risk stake $\Lambda_i = L_i(1 - g_i)$, users rationally specify the highest-stakes attributes first, while assistants prioritize asking on the highest-residual-stake attributes. The fundamental contract-theoretic mechanism—the substitution between upstream specification effort and automated clarification under misaligned asking friction—generalizes directly to the heterogeneous-loss setting.

A related limitation concerns identification of λ_A itself. Every bias-driven prediction (Corollary 2, Proposition 6, and predictions P2, P6, P7) is stated in terms of $\lambda_A > \mu_A c_Q$, but the paper does not currently propose a way to measure λ_A independent of the ask-rate behavior it is meant to explain; backing it out from observed policy alone is circular, and Corollary 3 sharpens

this into an exact non-identification result on the interior branch: a^{SE} depends on (μ_A, λ_A) only through their ratio, so a discounting-driven and a friction-driven leader with the same λ_A/μ_A are indistinguishable from pooling behavior alone. An independent handle—for instance, from a training objective or reward specification that explicitly penalizes turn count—would be needed before the bias-versus-heterogeneity predictions in Section 8 can be tested non-circularly. The two-type restriction in Model II avoids ironing/continuum-type complications at the cost of generality; a continuum version using the standard Mirrlees technique is the natural next step and would let κ^* from the first-best benchmark interact directly with the screening distortion. We also assume truthful revelation of specified attributes (no cheap-talk vagueness within what is said) and a single leader (no competition among assistants, which would itself be an interesting extension—competition plausibly disciplines bias directly, another testable channel).

A broader methodological lesson from this model concerns the hazards of importing classical contract-theoretic intuition directly into human–AI settings. In classical mechanism design (e.g., Baron–Myerson, Laffont–Tirole), the principal pays monetary transfers to extract surplus, so agent rent directly diminishes principal payoff. In human–AI interaction, the assistant’s objective directly incorporates (a discounted multiple of) user welfare. As a result, incentive compatibility can be free in the absence of bias (Proposition 5), participation constraints do not automatically bind at the bottom of the type distribution, and unbiased pooling naturally induces corner solutions rather than interior compromises (Corollary 5). Contract-theoretic treatments of AI interaction must therefore explicitly account for the non-monetary, partially aligned nature of the interaction rather than relying on analogies to cash-transfer mechanisms.

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